Fourier Series and PDEs Problem Sheet 1

1. (a) Consider the boundary value problem for $y(x)$ given by

$$\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} + y = 0 \quad \text{for} \quad 0 < x < L,$$

with $y(0) = 0$ and $y(L) = 0$, where L is a positive constant. The function $y(x) = 0$ is a solution. Is this solution unique, or not? How does the answer depend on L?

(b) Consider the initial value problem for $y(t)$ given by

$$\frac{\mathrm{d}y}{\mathrm{d}t} = 2y^{1/2} \quad \text{for} \quad t > 0,$$

with $y(0) = 0$. The function $y(t) = 0$ is a solution. Use separation of variables to find a nontrivial solution. Hence, write down a solution that vanishes if and only if $t \leq a$, where a is a positive constant.

2. (a) Use the power series representation of the function $1/(1-z)$ with $z = R\mathrm{e}^{\mathrm{i}\theta}$, where $R \in [0, 1)$ and $\theta \in \mathbb{R}$, to establish an example of a convergent Fourier sine series.

(b) Assuming that the real constants $a_1, a_2, \ldots$ and $b_1, b_2, \ldots$ are such that the orders of summation and differentiation may be interchanged, show that

$$y(x,t) = \sum_{n=1}^{\infty} \big(a_n \cos(nt) + b_n \sin(nt)\big) \sin(nx)$$

satisfies the wave equation

$$\frac{\partial^2 y}{\partial t^2} = \frac{\partial^2 y}{\partial x^2} \quad \text{for} \quad 0 < x < \pi, \ t > 0,$$

and the boundary conditions $y(0,t) = 0$ and $y(\pi,t) = 0$ for $t > 0$. Hence, write down an infinite series solution for $y(x,t)$ that also satisfies the initial conditions

$$y(x,0) = \frac{R \sin x}{1 - 2R\cos x + R^2}, \quad \frac{\partial y}{\partial t}(x,0) = 0 \quad \text{for } 0 < x < \pi.$$

3. (a) Let f be a periodic function of period 2π. By making the substitution $s = 2\pi + t$ in one of the integrals in the identity

$$\int_{\alpha}^{\alpha+2\pi} f(s)\,\mathrm{d}s = \int_{\alpha}^{\pi} f(s)\,\mathrm{d}s + \int_{\pi}^{\alpha+2\pi} f(s)\,\mathrm{d}s,$$

show that, for any real α,

$$\int_{\alpha}^{\alpha+2\pi} f(s)\,\mathrm{d}s = \int_{-\pi}^{\pi} f(s)\,\mathrm{d}s.$$

(b) Let g be an even function with derivative g' and let

$$G(x) = \int_{0}^{x} g(s)\,\mathrm{d}s.$$

(i) Use the chain rule to show that g' is an odd function.

(ii) By making the substitution $s = -t$, show that G is an odd function.

(iii) Deduce that, for any real α,

$$\int_{-\alpha}^{\alpha} g(s)\,\mathrm{d}s = 2\int_{0}^{\alpha} g(s)\,\mathrm{d}s.$$

(c) Show that if h is an odd function then, for any real α,

$$\int_{-\alpha}^{\alpha} h(s)\,\mathrm{d}s = 0.$$

4. Let $f : [0, \pi] \to \mathbb{R}$.

(a) Let f_e be a periodic function of period 2π satisfying

$$f_e(x) = \begin{cases} f(x) & \text{for } 0 \leqslant x \leqslant \pi, \\ f(-x) & \text{for } -\pi < x < 0. \end{cases}$$

Show that f_e is an even function.

(b) Let f_o be a periodic function of period 2π satisfying

$$f_o(x) = \begin{cases} f(x) & \text{for } 0 \leqslant x \leqslant \pi, \\ -f(-x) & \text{for } -\pi < x < 0. \end{cases}$$

Under what conditions on f is f_o an odd function? Justify your answer.